

Ініціалізація генераторів даних та конвеєра аугментації...

Found 600 images belonging to 2 classes.

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Архітектура Inception-v3 успішно побудована пошарово.

Файл моделі 'my_custom_inception_model.h5' не знайдено. Початок навчання...

/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121:

UserWarning: Your `PyDataset` class should call `super().\_\_init\_\_(\*\*kwargs)` in its constructor. `\*\*kwargs` can include `workers`, `use\_multiprocessing`, `max\_queue\_size`. Do not pass these arguments to `fit()`, as they will be ignored.

```
self._warn_if_super_not_called()
```

Epoch 1/10

18/18  **215s** 11s/step - accuracy: 0.5798 - loss: 0.7033 - val_accuracy: 0.5208 - val_loss: 0.6925

Epoch 2/10

1/18  **2:50** 10s/step - accuracy: 0.6875 - loss: 0.6061

/usr/local/lib/python3.12/dist-packages/keras/src/trainers/epoch_iterator.py:116: UserWarning: Your input ran out of data; interrupting training. Make sure that your dataset or generator can generate at least `steps\_per\_epoch \* epochs` batches. You may need to use the `.repeat()` function when building your dataset.

```
self._interrupted_warning()
```

18/18  **24s** 826ms/step - accuracy: 0.6875 - loss: 0.6061 - val_accuracy: 0.5208 - val_loss: 0.6926

Epoch 3/10

18/18  **233s** 11s/step - accuracy: 0.5906 - loss: 0.6606 - val_accuracy: 0.5208 - val_loss: 0.7020

Epoch 4/10

18/18  **41s** 2s/step - accuracy: 0.9062 - loss: 0.4651 - val_accuracy: 0.5208 - val_loss: 0.7050

Epoch 5/10

18/18  **196s** 11s/step - accuracy: 0.6999 - loss: 0.5885 - val_accuracy: 0.5208 - val_loss: 0.7338

Epoch 6/10

18/18  **24s** 848ms/step - accuracy: 0.5938 - loss: 0.6365 - val_accuracy: 0.5208 - val_loss: 0.7348

Epoch 7/10

18/18 ————— **202s** 11s/step - accuracy: 0.6902 - loss: 0.5923 - val_accuracy: 0.5208
- val_loss: 0.7504

Epoch 8/10

18/18 ————— **25s** 859ms/step - accuracy: 0.8750 - loss: 0.4597 - val_accuracy:
0.5208 - val_loss: 0.7513

Epoch 9/10

18/18 ————— **197s** 11s/step - accuracy: 0.7183 - loss: 0.5645 - val_accuracy: 0.5208
- val_loss: 0.8603

Epoch 10/10

18/18 ————— **24s** 833ms/step - accuracy: 0.6562 - loss: 0.6357 - val_accuracy:
0.5208 - val_loss: 0.8645

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the
native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Модель успішно збережено у файл 'my_custom_inception_model.h5'.

Генерація прогнозів для валідаційної вибірки...

7/7 ————— **16s** 2s/step

--- Метрики якості (Accuracy, Precision, Recall, F1-Score) ---

	precision	recall	f1-score	support
cats	0.50	1.00	0.67	100
dogs	0.00	0.00	0.00	100
accuracy		0.50	200	
macro avg	0.25	0.50	0.33	200
weighted avg	0.25	0.50	0.33	200

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division`
parameter to control this behavior.

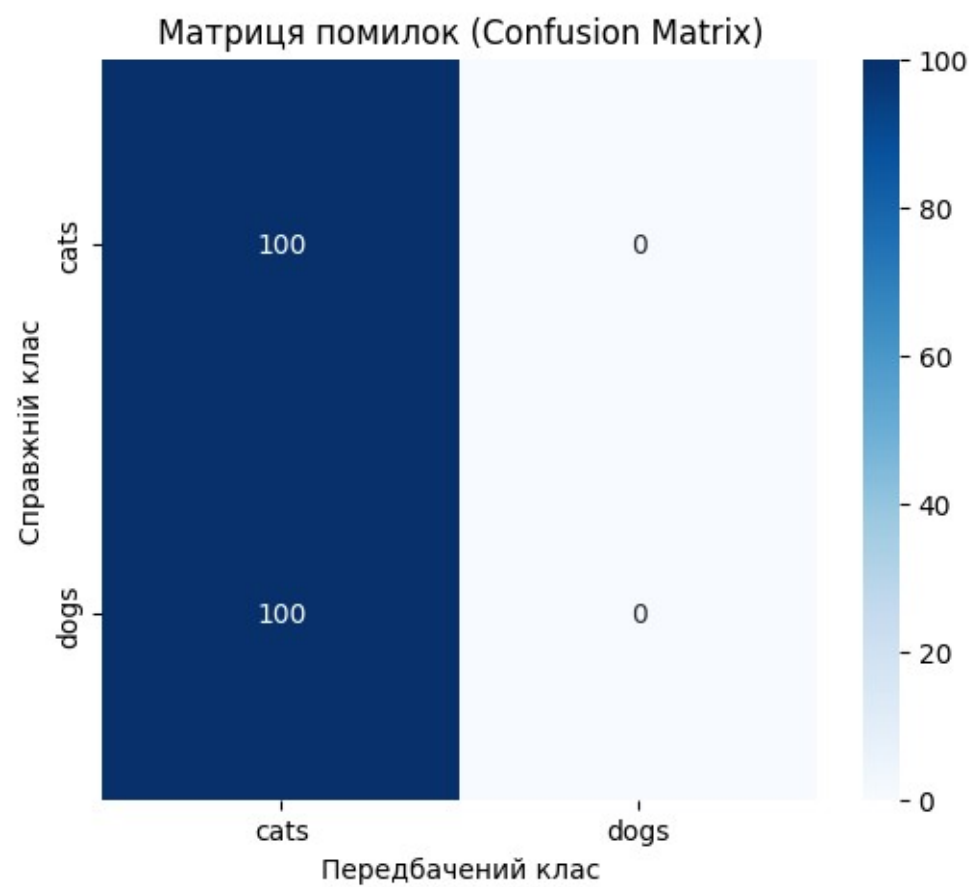
_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division`
parameter to control this behavior.

_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples. Use `zero\_division` parameter to control this behavior.

```
_warn_prf(average, modifier, f'{metric.capitalize()} is', len(result))
```



=== ПОРІВНЯННЯ ОБЧИСЛЮВАЛЬНОЇ СКЛАДНОСТІ ARCHITECTURES ===

1. Специфікація повної кастомної Inception-v3:

Model: "custom_inception_v3"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 299, 299, 3)	0	-
conv2d_22 (Conv2D)	(None, 149, 149, 32)	864	input_layer_1[0]...

batch_normalizatio...	(None, 149, 149,	128	conv2d_22[0][0]
(BatchNormalizatio...	32)		
activation_22	(None, 149, 149,	0	batch_normalizat...
(Activation)	32)		
conv2d_23 (Conv2D)	(None, 147, 147,	9,216	activation_22[0]...
	32)		
batch_normalizatio...	(None, 147, 147,	128	conv2d_23[0][0]
(BatchNormalizatio...	32)		
activation_23	(None, 147, 147,	0	batch_normalizat...
(Activation)	32)		
conv2d_24 (Conv2D)	(None, 147, 147,	18,432	activation_23[0]...
	64)		
batch_normalizatio...	(None, 147, 147,	256	conv2d_24[0][0]
(BatchNormalizatio...	64)		
activation_24	(None, 147, 147,	0	batch_normalizat...
(Activation)	64)		
max_pooling2d_2	(None, 73, 73,	0	activation_24[0]...
(MaxPooling2D)	64)		
conv2d_25 (Conv2D)	(None, 73, 73,	5,120	max_pooling2d_2[...
	80)		
batch_normalizatio...	(None, 73, 73,	320	conv2d_25[0][0]
(BatchNormalizatio...	80)		

activation_25	(None, 73, 73,	0	batch_normalizat...
(Activation)	80)		
conv2d_26 (Conv2D)	(None, 71, 71,	138,240	activation_25[0]...
	192)		
batch_normalizatio...	(None, 71, 71,	768	conv2d_26[0][0]
(BatchNormalizatio...	192)		
activation_26	(None, 71, 71,	0	batch_normalizat...
(Activation)	192)		
max_pooling2d_3	(None, 35, 35,	0	activation_26[0]...
(MaxPooling2D)	192)		
conv2d_30 (Conv2D)	(None, 35, 35,	12,288	max_pooling2d_3[...
	64)		
batch_normalizatio...	(None, 35, 35,	256	conv2d_30[0][0]
(BatchNormalizatio...	64)		
activation_30	(None, 35, 35,	0	batch_normalizat...
(Activation)	64)		
conv2d_28 (Conv2D)	(None, 35, 35,	9,216	max_pooling2d_3[...
	48)		
conv2d_31 (Conv2D)	(None, 35, 35,	55,296	activation_30[0]...
	96)		
batch_normalizatio...	(None, 35, 35,	192	conv2d_28[0][0]

(BatchNormalizatio...	48)		
batch_normalizatio...	(None, 35, 35,	384	conv2d_31[0][0]
(BatchNormalizatio...	96)		
activation_28	(None, 35, 35,	0	batch_normalizat...
(Activation)	48)		
activation_31	(None, 35, 35,	0	batch_normalizat...
(Activation)	96)		
average_pooling2d_3	(None, 35, 35,	0	max_pooling2d_3[...
(AveragePooling2D)	192)		
conv2d_27 (Conv2D)	(None, 35, 35,	12,288	max_pooling2d_3[...
	64)		
conv2d_29 (Conv2D)	(None, 35, 35,	27,648	activation_28[0]...
	64)		
conv2d_32 (Conv2D)	(None, 35, 35,	82,944	activation_31[0]...
	96)		
conv2d_33 (Conv2D)	(None, 35, 35,	6,144	average_pooling2...
	32)		
batch_normalizatio...	(None, 35, 35,	256	conv2d_27[0][0]
(BatchNormalizatio...	64)		
batch_normalizatio...	(None, 35, 35,	256	conv2d_29[0][0]
(BatchNormalizatio...	64)		

batch_normalizatio...	(None, 35, 35,	384	conv2d_32[0][0]
(BatchNormalizatio...	96)		
batch_normalizatio...	(None, 35, 35,	128	conv2d_33[0][0]
(BatchNormalizatio...	32)		
activation_27	(None, 35, 35,	0	batch_normalizat...
(Activation)	64)		
activation_29	(None, 35, 35,	0	batch_normalizat...
(Activation)	64)		
activation_32	(None, 35, 35,	0	batch_normalizat...
(Activation)	96)		
activation_33	(None, 35, 35,	0	batch_normalizat...
(Activation)	32)		
concatenate_4	(None, 35, 35,	0	activation_27[0]...
(Concatenate)	256)		activation_29[0]...
			activation_32[0]...
			activation_33[0]...
conv2d_35 (Conv2D)	(None, 35, 35,	32,768	concatenate_4[0]...
	128)		
batch_normalizatio...	(None, 35, 35,	512	conv2d_35[0][0]
(BatchNormalizatio...	128)		
activation_35	(None, 35, 35,	0	batch_normalizat...
(Activation)	128)		

conv2d_36 (Conv2D)	(None, 35, 35, 128)	114,688	activation_35[0]...
batch_normalizatio...	(None, 35, 35, 128)	512	conv2d_36[0][0]
activation_36	(None, 35, 35, 128)	0	batch_normalizat...
average_pooling2d_4	(None, 35, 35, 256)	0	concatenate_4[0]...
conv2d_34 (Conv2D)	(None, 35, 35, 192)	49,152	concatenate_4[0]...
conv2d_37 (Conv2D)	(None, 35, 35, 192)	172,032	activation_36[0]...
conv2d_38 (Conv2D)	(None, 35, 35, 192)	49,152	average_pooling2...
batch_normalizatio...	(None, 35, 35, 192)	768	conv2d_34[0][0]
batch_normalizatio...	(None, 35, 35, 192)	768	conv2d_37[0][0]
batch_normalizatio...	(None, 35, 35, 192)	768	conv2d_38[0][0]
activation_34	(None, 35, 35, 192)	0	batch_normalizat...

activation_37	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
activation_38	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
concatenate_5	(None, 35, 35,	0	activation_34[0]...
(Concatenate)	576)		activation_37[0]...
			activation_38[0]...
conv2d_40 (Conv2D)	(None, 35, 35,	221,184	concatenate_5[0]...
	384)		
batch_normalizatio...	(None, 35, 35,	1,536	conv2d_40[0][0]
(BatchNormalizatio...	384)		
activation_40	(None, 35, 35,	0	batch_normalizat...
(Activation)	384)		
conv2d_41 (Conv2D)	(None, 35, 35,	442,368	activation_40[0]...
	384)		
conv2d_42 (Conv2D)	(None, 35, 35,	442,368	activation_40[0]...
	384)		
average_pooling2d_5	(None, 35, 35,	0	concatenate_5[0]...
(AveragePooling2D)	576)		
conv2d_39 (Conv2D)	(None, 35, 35,	184,320	concatenate_5[0]...
	320)		

batch_normalizatio...	(None, 35, 35,	1,536	conv2d_41[0][0]
(BatchNormalizatio...	384)		
batch_normalizatio...	(None, 35, 35,	1,536	conv2d_42[0][0]
(BatchNormalizatio...	384)		
conv2d_43 (Conv2D)	(None, 35, 35,	110,592	average_pooling2...
	192)		
batch_normalizatio...	(None, 35, 35,	1,280	conv2d_39[0][0]
(BatchNormalizatio...	320)		
activation_41	(None, 35, 35,	0	batch_normalizat...
(Activation)	384)		
activation_42	(None, 35, 35,	0	batch_normalizat...
(Activation)	384)		
batch_normalizatio...	(None, 35, 35,	768	conv2d_43[0][0]
(BatchNormalizatio...	192)		
activation_39	(None, 35, 35,	0	batch_normalizat...
(Activation)	320)		
concatenate_6	(None, 35, 35,	0	activation_41[0]...
(Concatenate)	768)		activation_42[0]...
activation_43	(None, 35, 35,	0	batch_normalizat...
(Activation)	192)		
concatenate_7	(None, 35, 35,	0	activation_39[0]...
(Concatenate)	1280)		concatenate_6[0]...

			activation_43[0]...
global_average_poo...	(None, 1280)	0	concatenate_7[0]...
(GlobalAveragePool...			
dense_2 (Dense)	(None, 1024)	1,311,744	global_average_p...
dropout_1 (Dropout)	(None, 1024)	0	dense_2[0][0]
dense_3 (Dense)	(None, 1)	1,025	dropout_1[0][0]

Total params: 10,554,149 (40.26 MB)

Trainable params: 3,515,809 (13.41 MB)

Non-trainable params: 6,720 (26.25 KB)

Optimizer params: 7,031,620 (26.82 MB)

2. Специфікація оптимізованої Mini-Inception:

Model: "mini_inception"

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 150, 150, 3)	0	-
conv2d_44 (Conv2D)	(None, 75, 75, 16)	432	input_layer_2[0]...
batch_normalizatio...	(None, 75, 75, 16)	64	conv2d_44[0][0]
activation_44 (Activation)	(None, 75, 75, 16)	0	batch_normalizat...

max_pooling2d_4	(None, 37, 37,	0	activation_44[0]...
(MaxPooling2D)	16)		
conv2d_46 (Conv2D)	(None, 37, 37,	256	max_pooling2d_4[...
	16)		
batch_normalizatio...	(None, 37, 37,	64	conv2d_46[0][0]
(BatchNormalizatio...	16)		
activation_46	(None, 37, 37,	0	batch_normalizatio...
(Activation)	16)		
max_pooling2d_5	(None, 37, 37,	0	max_pooling2d_4[...
(MaxPooling2D)	16)		
conv2d_45 (Conv2D)	(None, 37, 37,	512	max_pooling2d_4[...
	32)		
conv2d_47 (Conv2D)	(None, 37, 37,	4,608	activation_46[0]...
	32)		
conv2d_48 (Conv2D)	(None, 37, 37,	256	max_pooling2d_5[...
	16)		
batch_normalizatio...	(None, 37, 37,	128	conv2d_45[0][0]
(BatchNormalizatio...	32)		
batch_normalizatio...	(None, 37, 37,	128	conv2d_47[0][0]
(BatchNormalizatio...	32)		
batch_normalizatio...	(None, 37, 37,	64	conv2d_48[0][0]
(BatchNormalizatio...	16)		

activation_45	(None, 37, 37,	0	batch_normalizat...
(Activation)	32)		
activation_47	(None, 37, 37,	0	batch_normalizat...
(Activation)	32)		
activation_48	(None, 37, 37,	0	batch_normalizat...
(Activation)	16)		
concatenate_8	(None, 37, 37,	0	activation_45[0]...
(Concatenate)	80)		activation_47[0]...
			activation_48[0]...
global_average_poo...	(None, 80)	0	concatenate_8[0]...
(GlobalAveragePool...			
dense_4 (Dense)	(None, 1)	81	global_average_p...

Total params: 6,593 (25.75 KB)

Trainable params: 6,369 (24.88 KB)

Non-trainable params: 224 (896.00 B)

Висновок: Mini-Inception оперує значно меншою кількістю параметрів, зменшуючи використання оперативної пам'яті (VRAM) та прискорюючи крок навчання, проте може демонструвати нижчу узагальнюючу здатність на складних вибірках.